Linnea M. Wolniewicz

Ph.D. Candidate in Computer Science at the University of Hawai'i at Mānoa and National Science Foundation Fellow

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Education

University of Hawai'i at Mānoa *Honolulu, HI*

Ph.D. in Computer Science | M.Sc. in Computer Science (GPA 3.96) Aug. 2022 - May 2027 (expected)

Ph.D. Candidate since June 2026 in Dr. Peter Sadowski's Applied Machine Learning lab. Coursework: Machine Learning, Deep Learning, AI for Dynamic Systems, Random Processes, Complexity Analysis.

University of Colorado Boulder *Boulder, CO*

B.A. in Astrophysics and Minor in Music (Harp performance), GPA 3.86 August 2018 - May 2022

Coursework: Data Structures, Algorithms, Scientific Programming, Classical Mechanics, Quantum Mechanics, Electricity & Magnetism, Astrophysics.

Technical Skills

Programming Languages Python, C++, bash, LaTeX. *Working knowledge:* SQL, Scala, JavaScript, HTML/CSS

Data Science PyTorch, TensorFlow, HPC, Pandas, Matplotlib, SLURM, GCP. *Working knowledge:* MATLAB, Mathematica

Employment and Research

University of Hawai'i at Mānoa *Honolulu, HI*

Graduate Research Assistant August 2022 - present, full-time

Research **probabilistic machine learning** with Dr. Peter Sadowski, applying **Gaussian Processes** in **Python** for anomaly detection and accelerating **simulation-based inference** via **neural likelihood models** [2].

NVIDIA *Santa Clara, CA*

SW-GPU Intern, PhysicsNeMo / Earth-2 July 2026 - October 2026, full-time

Researched deep learning applications for climate science, developing generative satellite and radar weather models.

NASA Jet Propulsion Laboratory, Caltech *Pasadena, CA*

Research Intern, MLIA March 2026 - June 2026, full-time

Optimized ionospheric deep learning architectures to reduce runtime and built evaluation pipelines across geomagnetic storm conditions [1].

Frontier Development Lab *Remote*

Researcher at HelioLab June 2025 - October 2025, full-time

Developed a novel dataset and deep learning models in **PyTorch** to forecast ionospheric dynamics, presented at NeurIPS [3, 4].

Striveworks, Inc *Austin, TX*

Data Science Intern May 2023 - August 2023, full-time

Researched **large language models** for NLP tasks, evaluating parameter-efficient fine-tuning strategies such as LoRA.

Laboratory for Atmospheric and Space Physics *Boulder, CO*

Undergraduate Research Assistant August 2020 - July 2022, part-time

Implemented K-means and **CNNs** in **PyTorch** to segment solar polar coronal holes [6].

MIT Haystack Observatory *Westford, MA*

Research Experience for Undergraduates May 2021 - August 2021, full-time

Applied GMMs and **CNNs** to detect ocean wave events in seismic data from Antarctica's Ross Ice Shelf.

Institute for Astronomy at the University of Hawai'i at Mānoa *Honolulu, HI*

Research Experience for Undergraduates May 2020 - August 2020, full-time

Analyzed the entire Kepler Space Telescope dataset using **Python** statistical tools such as **Pandas** to evaluate the biases in its selection function [5].

Publications

[1] **Wolniewicz, L. M.**, Rebbapragada, U. D., McGranaghan, R. M., Verkhoglyadova, O., Crichton, D., ..., & Sadowski, P., 2026. "Ionospheric Forecasts Across Geomagnetic Conditions: Comparing Vision, Graph, and Recurrent Models"[Workshop paper]. *NeurIPS*, Advances in Representation Learning for Earth Observation workshop.

[2] **Wolniewicz, L. M.**, Sadowski, P., Corti, C., 2026. "Neural Surrogate HMC: On Using Neural Likelihoods for Hamiltonian Monte Carlo in Simulation-Based Inference" *JGR: Machine Learning and Computation*, 3(5), e2025JH001164. doi:10.1029/2025JH001164. [Link]

[3] **Wolniewicz, L.M.**, Kelebek, H.S., Mestici, S., Vergalla, M.D., Acciarini, G., ... & Soboczenski, F., 2025. "Connecting the Dots: A Machine Learning Ready Dataset for Ionospheric Forecasting Models"[Workshop paper]. *NeurIPS*, Machine Learning for the Physical Sciences workshop. [Link]

[4] Kelebek, H.S., **Wolniewicz, L.M.**, Vergalla, M.D., Mestici, S., Acciarini, G., ... & Baydin, A.G., 2025. "Ioncast: A deep learning framework for forecasting ionospheric dynamics"[Workshop paper]. *NeurIPS*, Machine Learning for the Physical Sciences workshop. [Link]

[5] **Wolniewicz, L. M.**, Berger, T., Huber, D., 2021. "The Stars Kepler Missed: Investigating the Kepler Target Selection Function Using Gaia DR2" *The Astronomical Journal*, 161(5). [Link]

[6] Tiwari, A.J., Hu, A., Tremblay, B., Smith, B., **Wolniewicz, L.M.**, ... & Samanta, T., 2020. "SEARCH: SEgmentation of polAR Coronal Holes"[Workshop paper]. *NeurIPS*, Machine Learning for the Physical Sciences workshop. [Link]

Awards and Honors

Jan. & May 2024 Catalyst Award for Science Advancement (CASA) Grant for Outreach with Graduate Women in Science Hawai'i

April 2022 National Science Foundation Graduate Fellowship